Sparse-Aware Spatio-Temporal Demand Forecasting

1 Input Representation

1.1 Temporal Context Embedding

We model calendar periodicity using learnable embedding tables.

Day-of-week embedding. Let $\mathrm{dow}(t) \in \{1, \ldots, 7\}$ denote the day of week at time t.

$$\mathbf{E}_{\mathrm{dow}} \in \mathbb{R}^{7 \times d_t}, \quad d_t = 16, \quad \mathbf{u}_t^{\mathrm{dow}} = \mathbf{E}_{\mathrm{dow}}[\mathrm{dow}(t)].$$

Hour-of-day embedding. Let $\mathrm{hod}(t) \in \{0, \ldots, 23\}$ denote the hour of day.

$$\mathbf{E}_{\mathrm{hod}} \in \mathbb{R}^{24 \times d_t}, \quad \mathbf{u}_t^{\mathrm{hod}} = \mathbf{E}_{\mathrm{hod}}[\mathrm{hod}(t)].$$

Holiday embedding. We introduce a learnable holiday embedding vector:

$$\mathbf{e}_{\mathrm{hol}} \in \mathbb{R}^{d_t}.$$

Temporal context representation.

$$\mathbf{u}_t^{\mathrm{time}} = \mathbf{u}_t^{\mathrm{dow}} + \mathbf{u}_t^{\mathrm{hod}} + \mathbb{I}(\mathrm{holiday}(t)) \cdot \mathbf{e}_{\mathrm{hol}}.$$

1.2 Region Features

Each region corresponds to an irregular spatial area. Static features describe functional and geographic characteristics and remain constant over time.

Land-use composition. Let C_{land} denote the number of land-use categories.

$$\mathbf{x}_r^{\mathrm{land}} \in \mathbb{R}^{C_{\mathrm{land}}}, \quad \sum_{k=1}^{C_{\mathrm{land}}} x_{r,k}^{\mathrm{land}} = 1.$$

Example:

$$\mathbf{x}_r^{\mathrm{land}} = [0.6 \text{ (res)}, 0.3 \text{ (comm)}, 0.1 \text{ (office)}, 0, 0].$$

Geographic attributes. Geographic features encode region location and size:

$$\mathbf{x}_r^{\mathrm{geo}} = [\mathrm{lat}_r, \mathrm{lon}_r, \log(1 + \mathrm{area}_r)].$$

Time-conditioned POI statistics with learnable importance. Let C_{poi} denote the number of POI categories. We use raw POI counts without explicit area normalization, since region area is separately encoded as a geographic attribute.

$$\mathbf{x}_r^{\text{poi}} \in \mathbb{R}_{\geq 0}^{C_{\text{poi}}}.$$

We decompose POI information into presence and magnitude components:

$$z_{r,c} \triangleq \mathbb{I}(x_{r,c}^{\text{poi}} > 0), \quad m_{r,c} \triangleq \log(1 + x_{r,c}^{\text{poi}}), \quad c = 1, \dots, C_{\text{poi}}.$$

To allow the impact of POI categories to vary with time, we introduce a lightweight gating mechanism conditioned on the temporal embedding. For each POI category c, we define:

$$\gamma_{c,t} = \sigma(\mathbf{w}_c^{(\gamma)\top} \mathbf{u}_t^{\text{time}} + b_c^{(\gamma)}), \quad \gamma_{c,t} \in (0,1).$$

The time-conditioned POI feature is given by:

$$\tilde{x}_{r,c,t}^{\text{poi}} = \gamma_{c,t} \left(w_c^{(z)} z_{r,c} + w_c^{(m)} m_{r,c} \right),$$

$$\tilde{\mathbf{x}}_{r,t}^{\text{poi}} = [\tilde{x}_{r,1,t}^{\text{poi}}, \dots, \tilde{x}_{r,C_{\text{poi}},t}^{\text{poi}}],$$

where $w_c^{(z)}$ and $w_c^{(m)}$ are respectively defined as softplus($\theta_c^{(z)}$) and softplus($\theta_c^{(m)}$), with $\theta_c^{(z)}, \theta_c^{(m)} \in \mathbb{R}$ being learnable scalar parameters.

Region representation.

$$\mathbf{u}_{r,t}^{\text{reg}} = f_{\text{reg}}(\mathbf{x}_r^{\text{land}} \| \tilde{\mathbf{x}}_{r,t}^{\text{poi}} \| \mathbf{x}_r^{\text{geo}}), \quad \mathbf{u}_{r,t}^{\text{reg}} \in \mathbb{R}^{d_r},\ d_r = 32.$$

1.3 Demand Features (Local Lag Encoding)

Lag window. Let ℓ denote the local lag window length:

$$\mathbf{y}_{r,t}^{(\ell)} = [y_{r,t-\ell+1}, \dots, y_{r,t}].$$

Boundary handling. For k such that $t - k < 0$:

$$y_{r,t-k} = 0, \quad \xi_{r,t,k} = \mathbb{I}(t - k \geq 0),$$

$$\xi_{r,t}^{(\ell)} = [\xi_{r,t,\ell-1}, \dots, \xi_{r,t,0}].$$

Sparsity descriptors.

$$c_{r,t} = \sum_{i=0}^{\ell-1} \mathbb{I}(y_{r,t-i} > 0),$$

$$\Delta t_{r,t}^{\text{last}} = \min\bigl(t - \max\{\tau \leq t : y_{r,\tau} > 0\}, \Delta_{\max}\bigr).$$

Demand feature representation.

$$\mathbf{u}_{r,t}^{\text{dem}} = f_{\text{dem}}(\mathbf{y}_{r,t}^{(\ell)} \| \xi_{r,t}^{(\ell)} \| c_{r,t} \| \Delta t_{r,t}^{\text{last}}), \quad \mathbf{u}_{r,t}^{\text{dem}} \in \mathbb{R}^{d_d},\ d_d = 32.$$

1.4 Initial Region-Time Embedding

$$\mathbf{e}_{r,t} = \phi\Big(\mathbf{W}_e[\mathbf{u}_{r,t}^{\text{reg}} \| \mathbf{u}_{r,t}^{\text{dem}} \| \mathbf{u}_t^{\text{time}}] + \mathbf{b}_e\Big), \quad \mathbf{e}_{r,t} \in \mathbb{R}^{d_e}, d_e = 64.$$

2 Spatio-Temporal State Update

The region-specific hidden state in our RNN model is updated as

$$\mathbf{h}_{r,t} = \text{GRU}(\mathbf{h}_{r,t-1}, \mathbf{z}'_{r,t}), \quad \mathbf{h}_{r,t} \in \mathbb{R}^{d_h},\ d_h = 64,$$

where $\mathbf{h}_{r,0} = \mathbf{0}$ and $\mathbf{z}'_{r,t}$ is an attention aggregated representation computed as follows.

We define the effective region representation at time t by combining historical and instantaneous information using gated fusion.

Gated fusion.

$$\mathbf{g}_{r,t} = \sigma(\mathbf{W}_g[\mathbf{e}_{r,t} \| \mathbf{h}_{r,t-1}] + \mathbf{b}_g),$$

$$\mathbf{s}_{r,t} = \mathbf{g}_{r,t} \odot \mathbf{h}_{r,t-1} + (1 - \mathbf{g}_{r,t}) \odot \mathbf{e}_{r,t}.$$

At each time step, spatial interactions among regions are modeled via global attention over the fused representations $\{\mathbf{s}_{r,t}\}_r$.

Attention projections. With $H = 4$ heads and $d_h = 16$:

$$\mathbf{q}_{r,t}^{(h)} = \mathbf{W}_q^{(h)} \mathbf{s}_{r,t}, \quad \mathbf{k}_{j,t}^{(h)} = \mathbf{W}_k^{(h)} \mathbf{s}_{j,t}, \quad \mathbf{v}_{j,t}^{(h)} = \mathbf{W}_v^{(h)} \mathbf{s}_{j,t}.$$

Attention score.

$$a_{rj,t}^{(h)} = \frac{(\mathbf{q}_{r,t}^{(h)})^\top \mathbf{k}_{j,t}^{(h)}}{\sqrt{d_h}}.$$

Aggregation.

$$\alpha_{rj,t}^{(h)} = \frac{\exp(a_{rj,t}^{(h)})}{\sum_m \exp(a_{rm,t}^{(h)})}, \quad \mathbf{z}_{r,t}^{(h)} = \sum_j \alpha_{rj,t}^{(h)} \mathbf{v}_{j,t}^{(h)}.$$

$$\mathbf{z}_{r,t} = \mathbf{W}_o[\mathbf{z}_{r,t}^{(1)} \| \cdots \| \mathbf{z}_{r,t}^{(H)}].$$

Residual update.

$$\mathbf{z}'_{r,t} = \text{LayerNorm}(\mathbf{s}_{r,t} + \mathbf{z}_{r,t}).$$

3 Temporal Aggregation and Output

Given a temporal window $T_t = \{t - T + 1, \ldots, t\}$, we apply temporal attention over $\{\mathbf{h}_{r,\tau}\}_{\tau \in T_t}$.

Temporal attention.

$$\beta_{r,\tau} = \frac{\exp(\mathbf{v}_a^\top \tanh(\mathbf{W}_a \mathbf{h}_{r,\tau}))}{\sum_{k \in T_t} \exp(\mathbf{v}_a^\top \tanh(\mathbf{W}_a \mathbf{h}_{r,k}))},$$

$$\bar{\mathbf{h}}_r = \sum_{\tau \in T_t} \beta_{r,\tau} \mathbf{h}_{r,\tau}.$$

4 Sparse-Aware Output Heads

Event probability.

$$p_{r,t+1} = \sigma(\mathbf{w}_p^\top \bar{\mathbf{h}}_r + b_p).$$

Conditional magnitude.

$$\hat{y}^+_{r,t+1} = \mathrm{softplus}(\mathbf{w}_+^\top \bar{\mathbf{h}}_r + b_+).$$

Final prediction.

$$\hat{y}_{r,t+1} = p_{r,t+1} \cdot \hat{y}^+_{r,t+1}.$$

5 Loss Function

Ground-truth event indicator.

$$\delta_{r,t+1} = \mathbb{I}(y_{r,t+1} > 0).$$

Event loss.

$$\mathcal{L}_{\mathrm{evt}} = -\sum_{r,t} [\delta_{r,t+1} \log p_{r,t+1} + (1 - \delta_{r,t+1}) \log(1 - p_{r,t+1})].$$

Magnitude loss.

$$\mathcal{L}_{\mathrm{mag}} = \sum_{r,t} \delta_{r,t+1} \left(\frac{|\hat{y}^+_{r,t+1} - y_{r,t+1}|}{1 + y_{r,t+1}} + \lambda |\hat{y}^+_{r,t+1} - y_{r,t+1}| \right).$$

Total loss.

$$\mathcal{L} = \mathcal{L}_{\mathrm{evt}} + \mathcal{L}_{\mathrm{mag}}.$$